

Amazon SQS FIFO Queue

To Begin with the Lab

Summary of the Lab

In this lab, a **FIFO queue** was created in **Amazon SQS** to understand ordered message delivery and duplicate message handling. The queue name had to end with `.fifo`, so the queue was created as `test.fifo`. The video demonstrated how **content-based deduplication** affects whether SQS checks duplicate messages using the message body or using a **deduplication ID** provided by the producer. It also showed how the **deduplication scope** changes the duplicate-checking boundary between the entire queue and a specific message group. After that, messages were sent from the SQS console to verify how FIFO queue behavior changes under different settings, including the effect of changing the group ID, changing the deduplication ID, and toggling content-based deduplication. Finally, the queue was purged and polled to confirm the expected FIFO behavior.

Understanding Amazon SQS FIFO Queue

A **FIFO queue** in Amazon SQS exists when message order matters and duplicate handling must be controlled more strictly than in a standard queue. It keeps messages in order and uses **deduplication** to avoid processing the same message twice. If **content-based deduplication** is off, the producer must provide a deduplication ID; if it is on, SQS uses the message body to detect duplicates. The **deduplication scope** decides whether duplicates are checked across the whole queue or only within a message group.

Example:

A payment system sends a message like payment done into a FIFO queue. If content-based deduplication is off, the app must send a deduplication ID such as 1 or 2. If the same message body is sent again within the allowed deduplication window, SQS blocks it when deduplication scope is set to the queue or the same group, depending on the setting. That prevents duplicate payment processing.

Lab Steps

Step 1 – Create the FIFO Queue

- Sign in to the AWS Management Console.
- Open **Amazon SQS**.
- Click **Create queue**.
- Select **FIFO** queue type.
- Enter the queue name: `test.fifo`

Amazon SQS > **Queues** > **Create queue**

Details

Type
Choose the queue type for your application or cloud infrastructure.

☐ **Standard** [Info](#)
At-least-once delivery, message ordering isn't preserved

- At-least-once delivery
- Best-effort ordering

☒ **FIFO** [Info](#)
First-in-first-out delivery, message ordering is preserved

- First-in-first-out delivery
- Exactly-once processing

You can't change the queue type after you create a queue.

Name
test.fifo
A queue name is case-sensitive and can have up to 80 characters. You can use alphanumeric characters, hyphens (-), and underscores (_).

Configuration [Info](#)
Set the maximum message size, visibility to other consumers, and message retention.

Visibility timeout [Info](#)
30 Seconds
Should be between 0 seconds and 12 hours.

Message retention period [Info](#)
4 Days
Should be between 1 minute and 14 days.

Delivery delay [Info](#)
0 Seconds
Should be between 0 seconds and 15 minutes.

Maximum message size [Info](#)
1024 KiB
Should be between 1 KiB and 1024 KiB.

- Keep the default FIFO settings.
- Leave **Content-based deduplication** turned off for the first test.
- Keep **Deduplication scope** set to the queue for the first test.

Amazon SQS > **Queues** > **Create queue**

FIFO queue settings

☐ **Content-based deduplication**
When content-based deduplication is enabled, the message deduplication ID is optional.

☒ **High throughput FIFO queue (recommended)** [Info](#)
Configure your FIFO queue for maximum throughput.

High throughput for FIFO queues supports a higher number of transactions per second, per API.

Deduplication scope [Info](#)
Specify the scope of deduplication for a FIFO queue.

☒ **Queue**
Deduplicate all messages in a queue.

☐ **Message group**
Deduplicate all messages that belong to the same message group.

FIFO throughput limit [Info](#)
Specify how to apply the throughput limit on FIFO queue.

☒ **Per queue**
Apply the throughput limit at the queue level.

☐ **Per message group ID**
Apply the throughput limit at the message group level.

Encryption [Info](#)
Amazon SQS provides in-transit encryption by default. To add at-rest encryption to your queue, enable server-side encryption.

Server-side encryption

☐ Disabled

☒ **Enabled**

Encryption key type

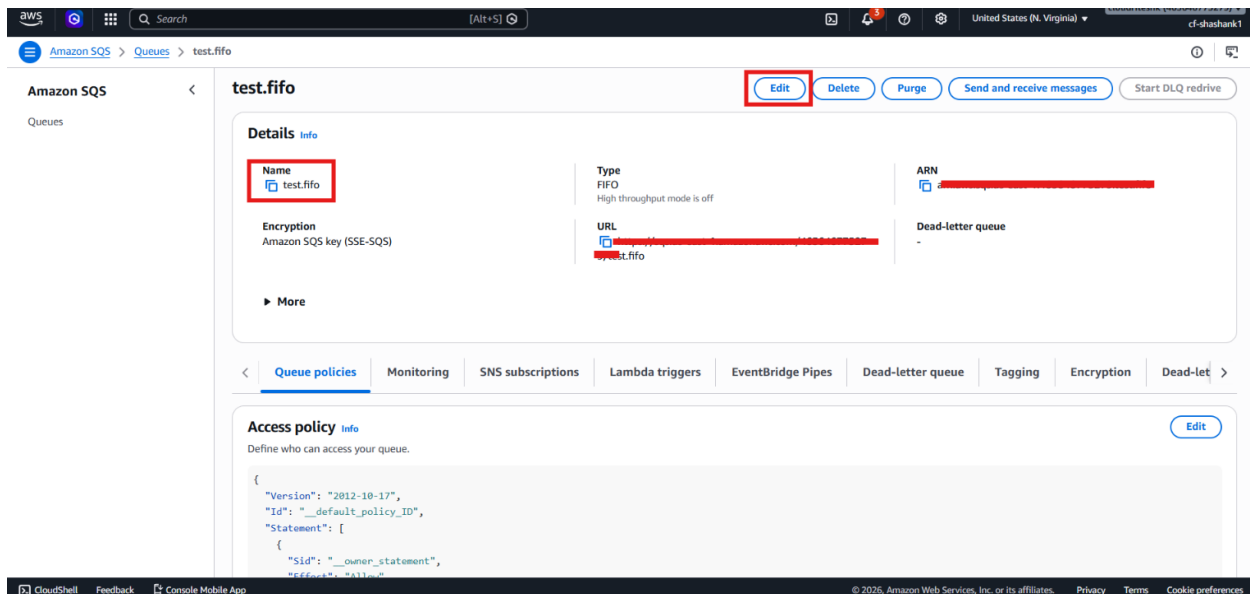
☒ **Amazon SQS key (SSE-SQS)**
An encryption key that Amazon SQS creates, manages, and uses for you.

☐ **AWS Key Management Service key (SSE-KMS)**
An encryption key protected by AWS Key Management Service (AWS KMS).

- Click **Create queue**.

Step 2 – Review FIFO Settings

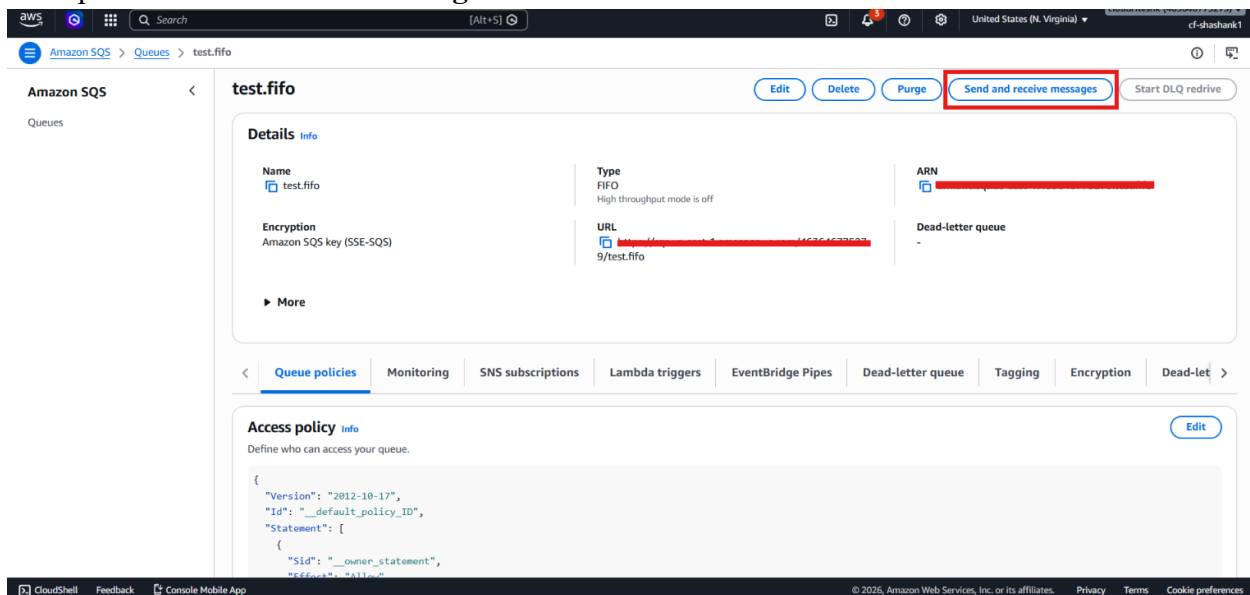
- Open the newly created queue.
- Click **Edit**.



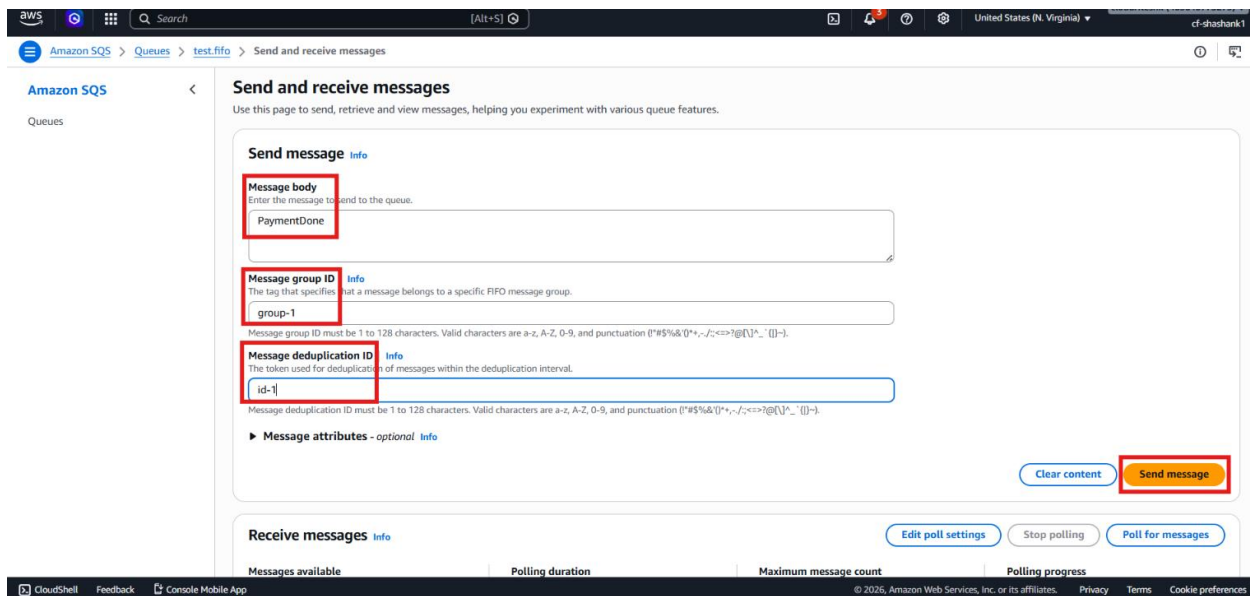
- Review **Content-based deduplication**.
- Review **Deduplication scope**.
- Confirm the current settings before testing.

Step 3 – Test FIFO with Content-Based Deduplication Off

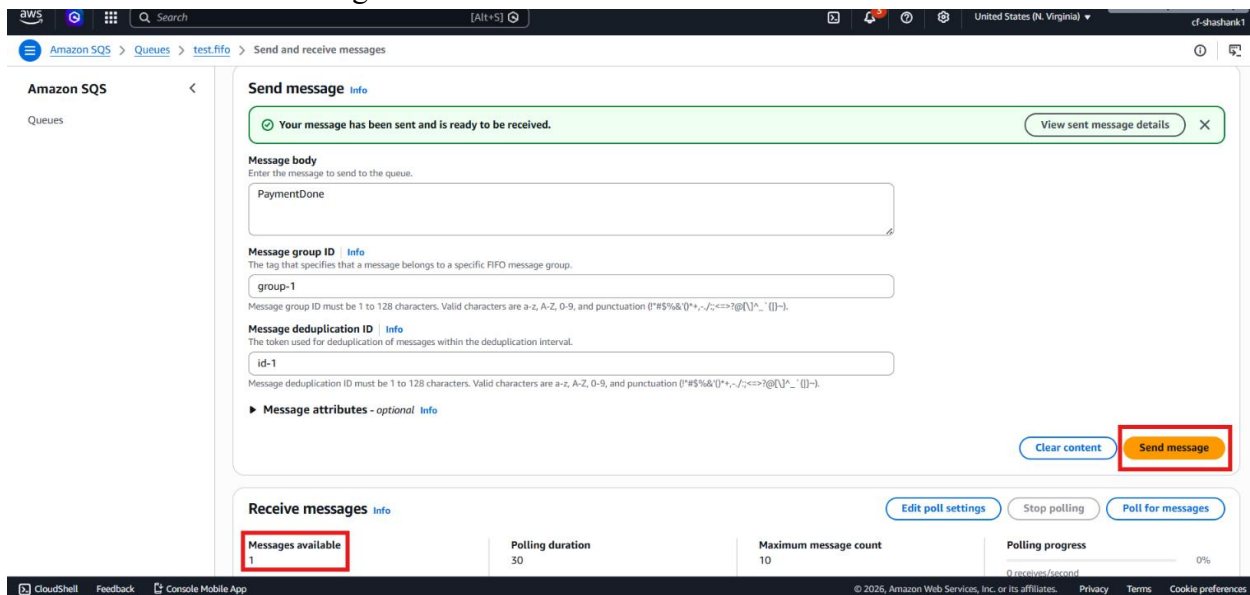
- Open **Send and receive messages**.



- Enter a message body: **PaymentDone**
- Enter a **Message group ID**: **group-1**
- Enter a **Message deduplication ID**: **id-1**
- Click **Send message**.

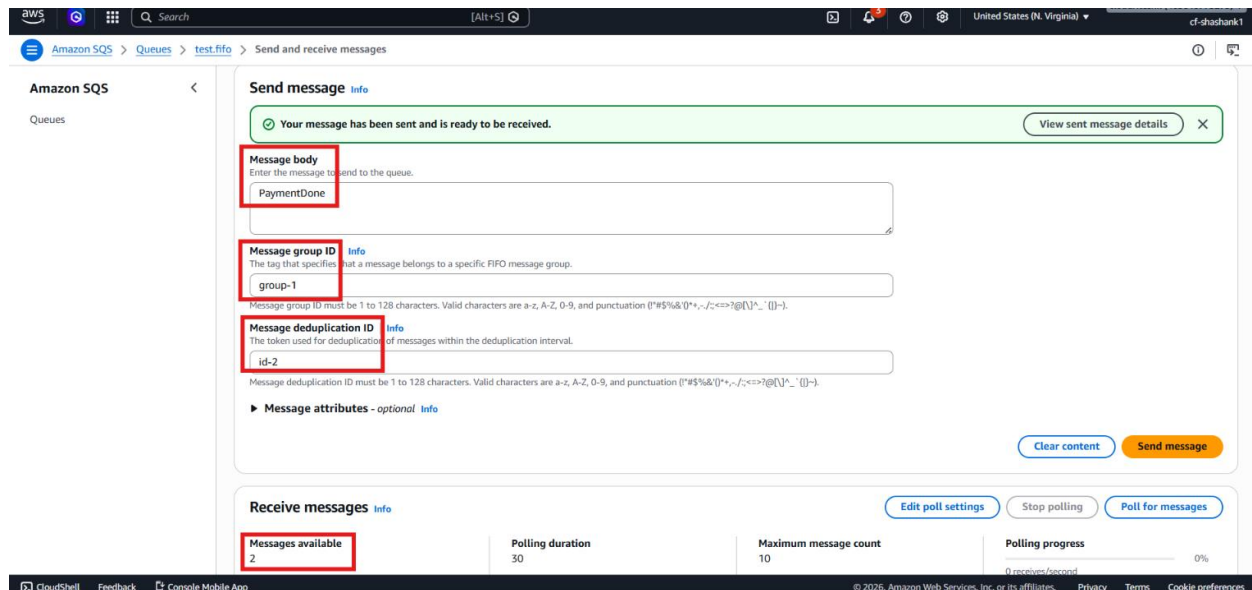


- Confirm that the message count increases to 1.

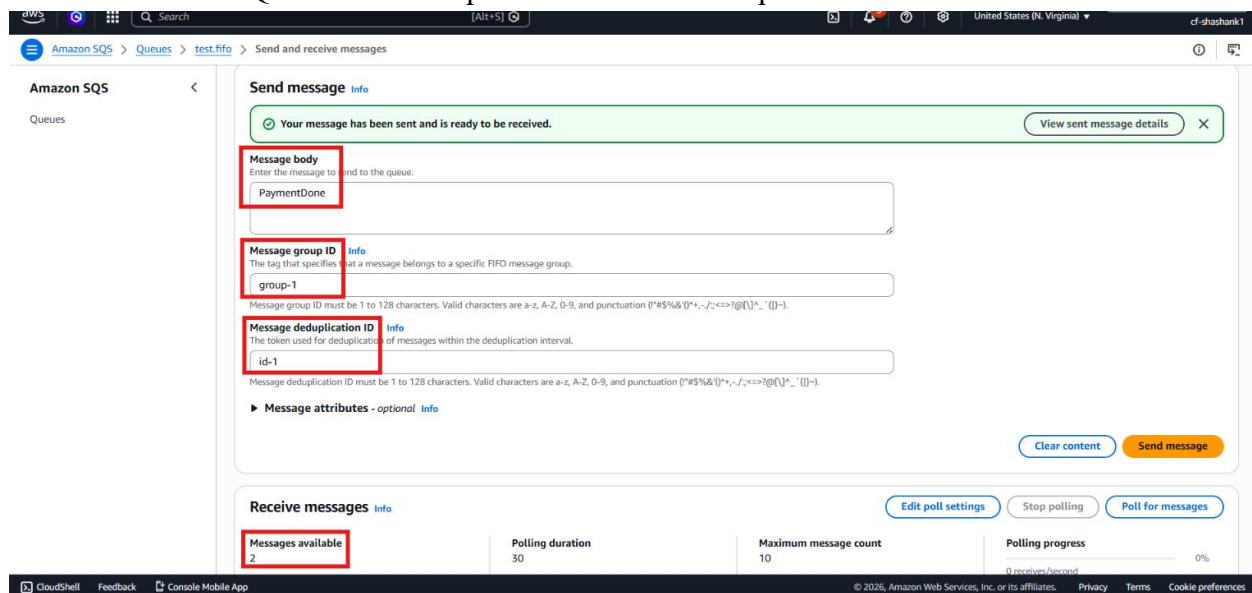


Step 4 – Test Duplicate Detection Using Deduplication ID

- Keep the same message body payment done.
- Keep the same message group ID group-1.
- Change the deduplication ID: id-2
- Click **Send message** again.
- Confirm that SQS accepts the message because the deduplication ID is different.

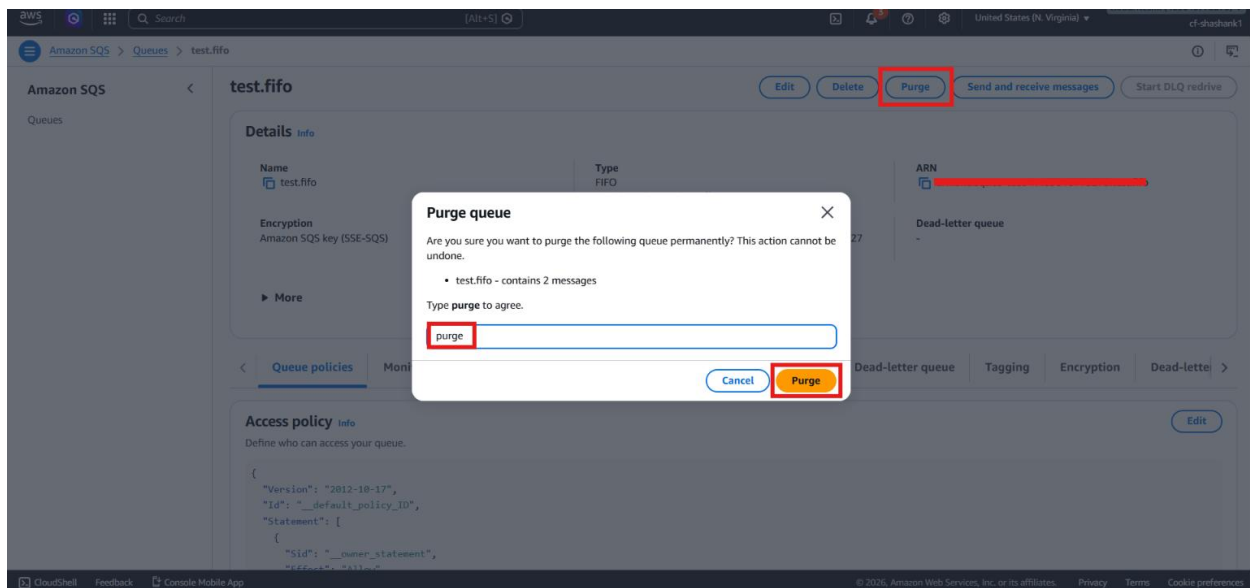


- Try sending the same message again with deduplication ID 1.
- Confirm that SQS blocks the duplicate within the deduplication window.

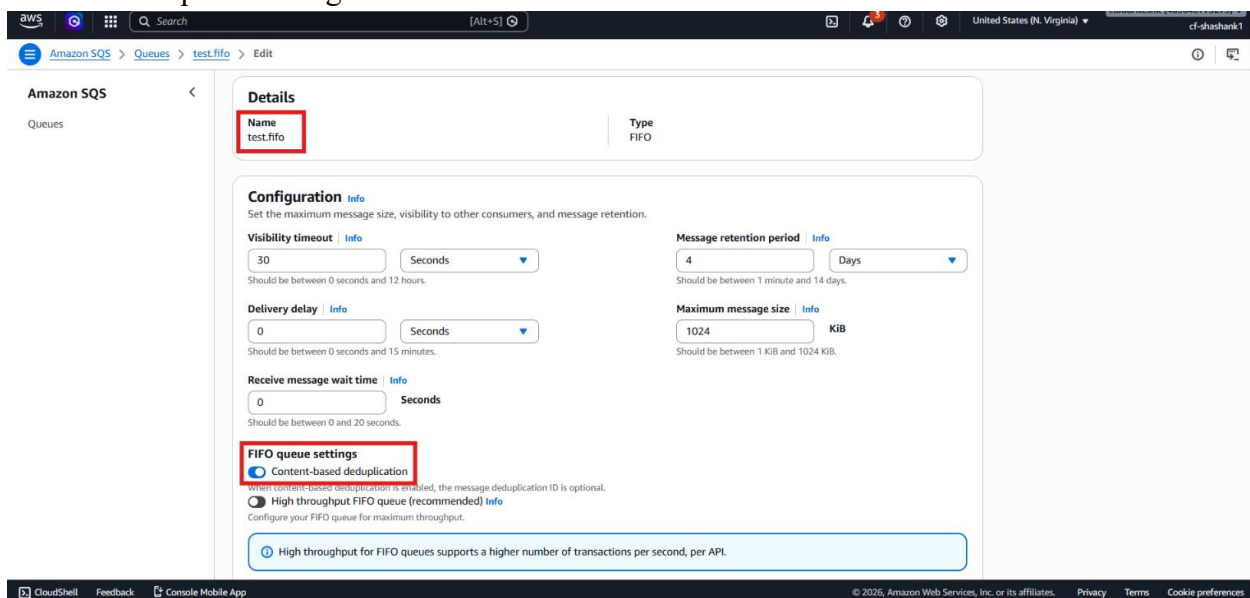


Step 5 – Clear the Queue and Enable Content-Based Deduplication

- Click **Purge** to remove all messages from the queue.



- Click on **Edit**.
- Turn **Content-based deduplication** on.
- Save the queue settings.



- Confirm that the deduplication ID becomes optional.

Step 6 – Test FIFO with Content-Based Deduplication On

- Open **Send and receive messages**.
- Enter the same message body again: **PaymentDone**
- Enter the same message group ID: **group-1**.
- Leave the deduplication ID blank.
- Click **Send message**.

Amazon SQS > Queues > test.fifo > Send and receive messages

Send message [Info](#)

✓ Your message has been sent and is ready to be received. [View sent message details](#) ✕

Message body [Info](#)
Enter the message to send to the queue.
PaymentDone

Message group ID [Info](#)
The tag that specifies that a message belongs to a specific FIFO message group.
group-1
Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[\]^_`{|}~]).

Message deduplication ID - optional [Info](#)
The token used for deduplication of messages within the deduplication interval.
Message deduplication ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[\]^_`{|}~]).

► **Message attributes - optional** [Info](#)

[Clear content](#) [Send message](#)

Receive messages [Info](#)

[Edit poll settings](#) [Stop polling](#) [Poll for messages](#)

Messages available 1 **Polling duration** 30 **Maximum message count** 10 **Polling progress** 0 receives/second 0%

- Try sending the same message again.
- Confirm that the duplicate is not accepted because SQS now checks the message body.

Amazon SQS > Queues > test.fifo > Send and receive messages

Send message [Info](#)

✓ Your message has been sent and is ready to be received. [View sent message details](#) ✕

Message body [Info](#)
Enter the message to send to the queue.
PaymentDone

Message group ID [Info](#)
The tag that specifies that a message belongs to a specific FIFO message group.
group-2
Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[\]^_`{|}~]).

Message deduplication ID - optional [Info](#)
The token used for deduplication of messages within the deduplication interval.
Message deduplication ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[\]^_`{|}~]).

► **Message attributes - optional** [Info](#)

[Clear content](#) [Send message](#)

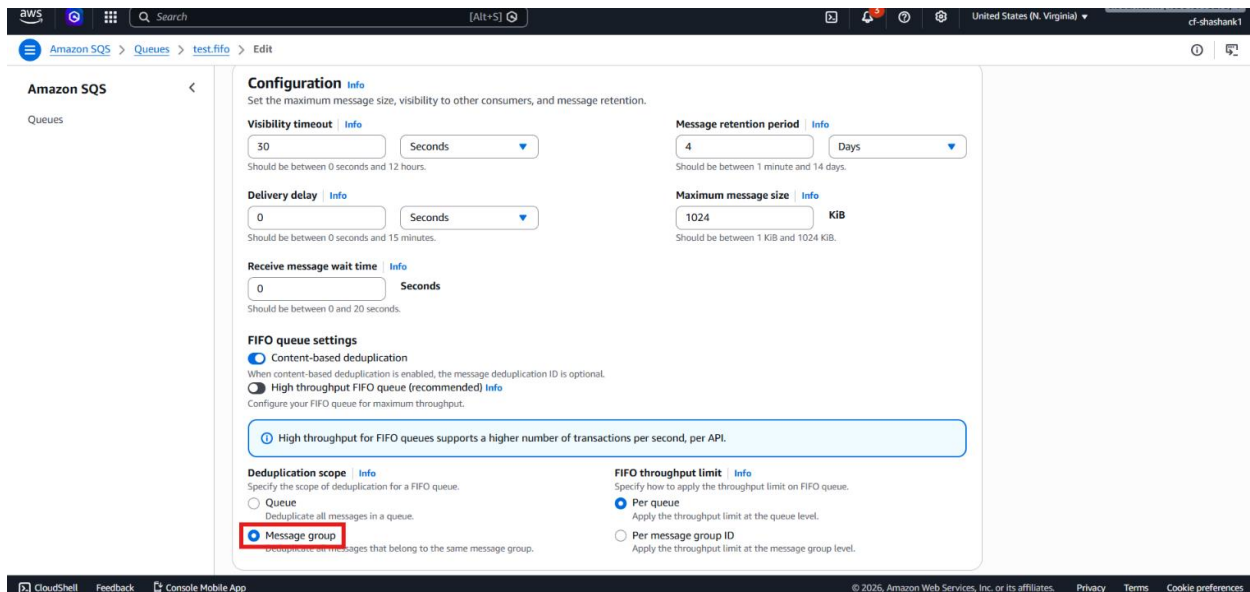
Receive messages [Info](#)

[Edit poll settings](#) [Stop polling](#) [Poll for messages](#)

Messages available 1 **Polling duration** 30 **Maximum message count** 10 **Polling progress** 0 receives/second 0%

Step 7 – Change the Deduplication Scope

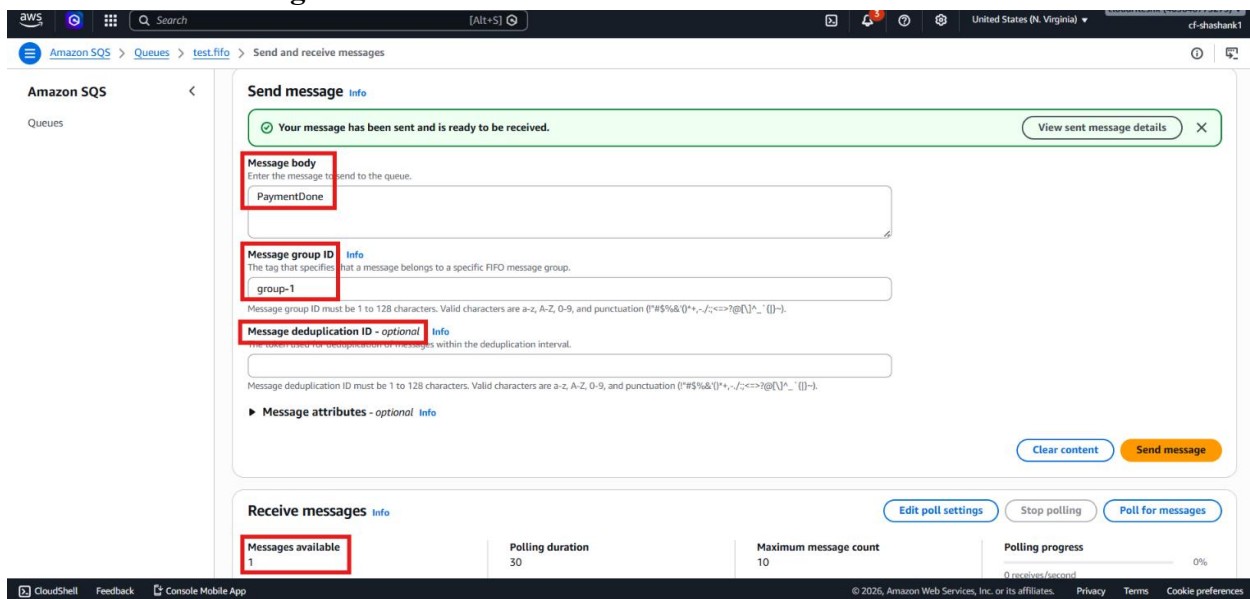
- Open **Edit** again.
- Change **Deduplication** scope from **Queue** to **Message group**.
- Save the changes.



- Purge the queue again if needed for a clean test.

Step 8 – Test Duplicate Behavior Across Groups

- Send the same message body payment done in group-1.
- Click **Send message**.



- Send the same message body again in the same group.
- Confirm that it is treated as a duplicate.

Send message [Info](#)

✓ Your message has been sent and is ready to be received. [View sent message details](#) ✕

Message body [Info](#)
Enter the message to send to the queue.
PaymentDone

Message group ID [Info](#)
The tag that specifies that a message belongs to a specific FIFO message group.
group-1
Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[^_`{|}~]).

Message deduplication ID - optional [Info](#)
The token used for deduplication of messages within the deduplication interval.
Message deduplication ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[^_`{|}~]).

► **Message attributes - optional** [Info](#)

[Clear content](#) [Send message](#)

Receive messages [Info](#)

[Edit poll settings](#) [Stop polling](#) [Poll for messages](#)

Messages available 1 **Polling duration** 30 **Maximum message count** 10 **Polling progress** 0 receives/second 0%

- Change the message group ID to a different value, such as group-2.
- Send the same message body again.

Send message [Info](#)

✓ Your message has been sent and is ready to be received. [View sent message details](#) ✕

Message body [Info](#)
Enter the message to send to the queue.
PaymentDone

Message group ID [Info](#)
The tag that specifies that a message belongs to a specific FIFO message group.
group-2
Message group ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[^_`{|}~]).

Message deduplication ID - optional [Info](#)
The token used for deduplication of messages within the deduplication interval.
Message deduplication ID must be 1 to 128 characters. Valid characters are a-z, A-Z, 0-9, and punctuation (["#\$%&'()*+,-./:;<=>?@[^_`{|}~]).

► **Message attributes - optional** [Info](#)

[Clear content](#) [Send message](#)

Receive messages [Info](#)

[Edit poll settings](#) [Stop polling](#) [Poll for messages](#)

Messages available 2 **Polling duration** 30 **Maximum message count** 10 **Polling progress** 0 receives/second 0%

- Confirm that the queue now accepts it because deduplication is limited to the message group.

Step 9 – Poll Messages from the Queue

- Open **Poll messages** or **Receive messages**.

The screenshot shows the Amazon SQS console interface. On the left, the navigation pane shows 'Amazon SQS' and 'Queues'. The main content area is titled 'Send and receive messages' for the queue 'test.fifo'. It includes a 'Message deduplication ID' section with a text input field and a 'Send message' button. Below this, the 'Receive messages' section displays 'Messages available: 2', 'Polling duration: 30', and 'Maximum message count: 10'. A 'Polling progress' bar shows 0% completion. A table of messages is shown with 0 messages. A red box highlights the 'Poll for messages' button.

- Confirm that the messages appear in the response.

The screenshot shows the Amazon SQS console interface after polling. The 'Receive messages' section now shows 'Messages available: 2' and 'Polling progress' at 20%. The table of messages now contains two entries, which are highlighted with a red box:

ID	Sent	Size	Receive count
034f9892-3bdd-4273-b43f-c2a8fbceda70	2026-07-02T13:23+05:30	11 bytes	1
aa76ede6-e973-4b58-a872-2f7375f65bb0	2026-07-02T13:24+05:30	11 bytes	1

- Observe that polling removes the message after it is received.
- Poll again and confirm that no messages remain.

Verification

- A FIFO queue named test.fifo was created successfully.
- The queue name ended with .fifo as required.
- Content-based deduplication behavior was tested.
- Deduplication ID behavior was tested when content-based deduplication was off.
- Deduplication scope was changed from queue-level to group-level.
- Duplicate message handling worked as expected.
- Polling confirmed FIFO queue message retrieval behavior.